

* Introduction:

→ With the rapid growth of IOT, smart home systems are becoming increasingly popular. These systems allow users to:

- Control appliances remotely
- Monitor home environments
- Improve safety and energy efficiency

→ It is widely used in such systems due to:

- Low power consumption.
- Low cost
- Mesh networking capability

→ However, while ZigBee enables efficient communication, it also introduces security challenges such as unauthorized access, key compromise, and network attacks.

* Technology used:

→ ZigBee Technology:

- Based on IEEE 802.15.4
- Low data rate (~250 Kbps)
- Supports mesh topology
- Used for device communication, sensors, and actuators.

→ Wi-Fi Technology:

- Used for remote access and mobile control
- High data rate

→ Home Gateway:

- Connects ZigBee and Wi-Fi networks
- Acts as central controller and protocol translator

→ Security Technologies:

- AES - 128 Encryption
- Trust Centre
- Install Codes
- Secure Remote password.

→ The system integrates:

- ZigBee based device network.
- Wi-Fi network
- Home gateway
- Security mechanisms

→ Goal:

- Provide remote control
- Ensure safety
- Protect against cyber threats.

→ Methodology:

- Device Setup: Sensors connected via ZigBee.
- Network Integration: ZigBee connected to Wi-Fi via gateway.
- User integration: Commands sent via mobile or remote.
- Security Validation: Checked using trust centre and encryption.
- Execution: Commands executed and feedback returned.

→ Security Testing:

- Planning
- Discovery
- Attack simulation
- Analysis.

→ System Architecture:

- ZigBee Network - Coordinators, routers, and devices
- Wi-fi Network - Mobile / Internet access
- Home Gateway - Central control unit
- Security Layer - Trust Centre, Encryption module

→ Data Flow: User → Wi Fi → Gateway → Security Check. → Zig Bee Device → Response.

→ Key Features:

- Remote home automation
- Low power communication
- Mesh networking
- Secure authentication
- Real-time monitoring

→ Security Challenge identified:

- Key management issues
- Unauthorized access.
- Denial-of-service attacks.
- Device compromise

→ Solutions Implemented.

- Install codes
- AES-128 encryption
- Centralized Trust centre
- Secure protocols
- Network monitoring

→ Advantages

- Cost-effective
- Energy efficient

→ Disadvantages

- Vulnerable to DOS attacks
- Depends on configurations

→ Conclusions

- ZigBee is effective for home automation. Integration with Wi-Fi improves usability and security mechanisms enhance reliability. However, security must be carefully managed.

→ Future scope:

- AI based intrusion detection.
- Blockchain security
- Voice control
- Cloud integration
- Advanced encryption
- Real-time threat monitoring.

Humy
22/3/26